Shenglong Chen

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Education

The University of British Columbia **Vancouver, BC**

Bachelor of Science, Major in Cognitive Systems (Artificial Intelligence) *Sep. 2022 – Nov. 2025*

- **Relevant Coursework:** Distributed Systems, Advanced Data Structures, Algorithm Design, Databases, Computer Systems, Artificial Intelligence, Statistical Modeling

Professional Experience

Simon Fraser University **Burnaby, BC**

Machine Learning Engineering Intern *May. 2025 – Oct. 2025*

- Engineered a high-performance ETL pipeline in Python for 10K+ financial records; implemented **vectorized preprocessing** and multiprocessing to achieve a **4x throughput increase** for downstream ML models.
- Designed relational schemas and optimized PostgreSQL queries through **B-tree indexing** and **EXPLAIN ANALYZE** profiling, reducing execution time by **35%** under high-concurrency stress tests.
- Developed a **Model Observability** dashboard using Python to monitor data drift and system health; implemented automated data validation in R to maintain **99.9% data integrity**.
- Applied **Test-Driven Development (TDD)**, increasing unit test coverage to **90%** and reducing production regressions by **20%**.

Key Technical Projects

Agent-Orchestrated Task Microservice: [**Python / Claude Code / MCP / Docker**] *Jan. 2026 – Apr. 2026*

- **Architected a distributed backend** by orchestrating **Claude Code** via **Git Worktrees** to coordinate parallel AI-assisted development of authentication and persistence layers.
- **Extended Agentic Context:** Utilized **Model Context Protocol (MCP)** to interface agents with PostgreSQL schemas, resulting in a **50% reduction** in schema-related bugs during autonomous refactoring.
- **Benchmarked System Reliability:** Benchmarked the system to handle **200+ concurrent requests** with sub-150ms tail latency by using agents to identify and resolve N+1 query bottlenecks in the data layer.
- **Automated Quality Assurance:** Provisioned **GitHub Actions** pipelines that utilized LLM-based agents for **Autonomous TDD**, improving test coverage for JWT-based authentication.

Distributed RESTful Task Microservice: [**Python / Flask / Docker / PostgreSQL**] *Jan. 2025 – Apr. 2025*

- Containerized backend services using **Docker Compose** , enabling one-command deployment across environments.
- **Implemented Secure Auth:** Integrated **JWT-based Authentication** and Bcrypt password hashing, successfully hardening the system against unauthorized access and credential-stuffing attacks during stress tests.
- **Built CI/CD Pipelines:** Leveraged **GitHub Actions** to automate unit testing and static code analysis, ensuring that 100% of merged code met production standards and passed all regression tests.
- **System Design:** Optimized read/write throughput via connection pooling and request validation logic, allowing the system to handle **200+ concurrent requests** with sub-150ms tail latency.

Event-Driven Data Processing Pipeline: [**Python / AWS Lambda / S3 / Docker / k6 / Locust**] *Jun. 2024 – Sep. 2024*

- Designed and implemented a serverless pipeline to process **1M+ events/day** using AWS Lambda and S3 with high availability
- Simulated **5,000+ concurrent users** using k6 and Locust, identifying database connection leaks and resource bottlenecks
- Reduced system latency by **40%** through connection pooling and optimized resource allocation
- Implemented structured logging and monitoring to detect anomalies and support debugging in distributed environments

AI Heuristic Solver Engine: [**Java / AI Algorithms / OOP**] *Jan. 2024 – Apr. 2024*

- Implemented a high-performance solver using heuristic search and pruning techniques to reduce search space
- Achieved **60% reduction in runtime** and **45% reduction in memory usage** through optimized data structures and caching
- Designed system using Strategy and Observer patterns to support extensibility and modular architecture

Technical Skills

Programming Languages: **Python** (Pandas, NumPy, Flask, FastAPI), **Java** (Spring Boot, JUnit 5, Swing), **SQL** (PostgreSQL, MySQL), R (Tidyverse), JavaScript, TypeScript

AI & Machine Learning: **PyTorch**, Scikit-Learn, **LLMs**, **RAG**, **Claude Code / Codex** (Code Generation, Debugging, Test Generation), Prompt Engineering, MCP (Tool Integration), Feature Engineering, Supervised/Unsupervised Learning

Cloud & DevOps: **AWS** (S3, EC2, Lambda), **Docker**, **CI/CD**, Linux/Unix, Git/GitHub, **TDD**, Unit Testing (PyTest, JUnit 5)

Software Engineering: **RESTful APIs**, Microservices, **Distributed Systems**, OOP, SOLID Principles, Design Patterns, Database Schema Design, Concurrency, Agile/Scrum

Data Engineering: **ETL Pipeline Design**, Multi-threaded Data Ingestion, **Multiprocessing**, Data Validation, Data Visualization (Matplotlib, Seaborn)

Certifications

AWS Certified Cloud Practitioner (Issued: May 2026), **AWS Solution Architect – Associate** (Issued: July 2026)